Robot Vision: Mock Exam 5

Chapters 2 to 12 · 90 minutes · no documents · 40 points

Answer on separate paper. Part A: one answer per question. Part B: two to four sentences. Part C: write the formula, then the numbers.

Part A: Multiple choice

22 points, about 35 min

A1 A network must predict three real values with no restriction on their range. Which output design is appropriate? (ch. 2)

(a) Three units with identity activation and a squared-error loss.
(b) A softmax over three units with cross-entropy.
(c) Three sigmoid units with binary cross-entropy.
(d) One sigmoid unit with squared error.

A2 What is the main purpose of momentum in SGD? (ch. 2)

(a) To switch off units randomly.
(b) To smooth oscillating updates and accelerate movement in consistent directions.
(c) To compute the Hessian of the loss.
(d) To reduce the number of parameters.

A3 A ReLU layer has fan-in M. What standard deviation does He initialization use for its weights? (ch. 3)

(a) $1/M$
(b) $2/M$
(c) $\sqrt{2/M}$
(d) $\sqrt{M/2}$

A4 Why is the learning rate often decreased during training? (ch. 3)

(a) To increase the number of parameters.
(b) Because a large rate is needed at the end.
(c) To make fast progress early and finer adjustments later.
(d) To make the loss noisier.

A5 What is the receptive field of a unit? (ch. 4)

(a) All the output maps of the layer.
(b) The region removed by pooling.
(c) The set of weights of its filter.
(d) The region of the input that can affect the output of that unit.

A6 Which statement about translation equivariance and invariance is correct? (ch. 4)

(a) Standard convolution guarantees invariance to any shift.
(b) Equivariance: shifting the input shifts the output; invariance: the output does not change.
(c) Equivariance: the output does not change; invariance: it shifts.
(d) Both mean that the output does not change.

A7 How does head-orientation estimation with a CNN differ from ordinary classification? (ch. 5)

(a) It predicts one class among three.
(b) It uses a softmax over angles.
(c) It predicts continuous values (roll, pitch, yaw) instead of class labels.
(d) It needs no training data.

A8 What is the main advantage of dilated convolution for segmentation? (ch. 5)

(a) It reduces the number of channels.
(b) A larger receptive field without extra weights and without downsampling.
(c) It removes the need for nonlinearities.
(d) It doubles the number of weights.

A9 In TCAV, what represents a concept, and what is the TCAV score? (ch. 6)

(a) A superpixel; the score is its Shapley value.
(b) A single neuron; the score is its maximum activation.
(c) A direction in a layer's activation space; the score is the fraction of class samples with a positive directional derivative along it.
(d) A filter; the score is its norm.

A10 A saliency map barely changes after the model weights are randomized. What does this show? (ch. 6)

(a) The model is very robust.
(b) The explanation does not really describe what the trained model has learned.
(c) The model has no parameters.
(d) The explanation is especially faithful.

A11 Why can LP-FT be better than direct fine-tuning? (ch. 7)

(a) Training the new head first, with the backbone frozen, prevents a random head from distorting the pretrained features.
(b) It trains the backbone first and the head never.
(c) It removes the pretrained features.
(d) It uses more labeled data.

A12 What is the difference between unsupervised domain adaptation and domain generalization? (ch. 7)

(a) Domain adaptation uses unlabeled target-domain data during training; domain generalization does not use target data.
(b) They are the same.
(c) Domain generalization uses labeled target data.
(d) Domain adaptation uses no source data.

A13 Which statement about out-of-distribution (OOD) inputs is correct? (ch. 8)

(a) A classifier can give a high softmax confidence to an OOD input.
(b) Softmax confidence is a reliable OOD detector.
(c) A classifier always gives a low confidence to OOD inputs.
(d) OOD inputs cannot be classified at all.

A14 What does quantization-aware training (QAT) do? (ch. 8)

(a) It quantizes a trained model without any retraining.
(b) It keeps float weights but simulates quantization in the forward pass during training.
(c) It trains a small student from a teacher.
(d) It removes whole filters.

A15 What does the recurrent part of RAFT do? (ch. 9)

(a) It segments the objects.
(b) It predicts the next frame.
(c) It processes the video frame by frame over time.
(d) It iteratively refines the optical-flow estimate for the same pair of images.

A16 Nodes are renumbered, with A and X permuted consistently. What should happen to the outputs? (ch. 9)

(a) Node-level outputs follow the permutation; graph-level outputs stay the same.
(b) Both stay the same.
(c) Both change unpredictably.
(d) Graph-level outputs follow the permutation; node outputs stay the same.

A17 Why is positional information added to the tokens of a Transformer? (ch. 10)

(a) Self-attention without it is permutation equivariant and does not know the order of the tokens.
(b) It makes the model recurrent.
(c) It replaces the value vectors.
(d) It speeds up the softmax.

A18 How do Swin and MaxViT make attention cheaper? (ch. 10)

(a) Swin uses grid attention; MaxViT uses shifted windows.
(b) Both remove attention.
(c) Swin uses local windows shifted between layers; MaxViT combines local block attention with sparse grid attention.
(d) Both compute full global attention at every layer.

A19 How is a large language model pre-trained? (ch. 11)

(a) By classifying manually labeled sentences.
(b) By reinforcement learning only.
(c) By predicting the next token on large amounts of text, which is self-supervised.
(d) By predicting images from text.

A20 How does CLIP classify an image "zero-shot"? (ch. 11)

(a) It generates a caption and searches it.
(b) It uses a softmax with one unit per ImageNet class.
(c) It compares the image embedding with the embeddings of texts like "a photo of a {class}" and picks the most similar.
(d) It fine-tunes on a few labeled images.

A21 What does pix2pix require, and what does its discriminator see? (ch. 12)

(a) No images; the discriminator sees noise.
(b) Unpaired images; the discriminator sees only the output.
(c) Paired images; the discriminator sees only the input.
(d) Paired images; the discriminator sees the input together with the output.

A22 What is the main advantage of a latent diffusion model? (ch. 12)

(a) It works only on pixels.
(b) It needs no denoising network.
(c) Diffusion runs in the smaller latent space of an autoencoder, which lowers the cost.
(d) It removes the need for any training data.

Part B: Short answers

12 points, 3 per question, about 35 min

B1 What does Adam combine, and why is it popular? (ch. 3)

B2 Why is the output of an FCN too coarse for segmentation, and how does U-Net fix it? (ch. 5)

B3 How does PointNet handle a set of points? Explain invariance and equivariance here. (ch. 9)

B4 What are the three stages of training a modern LLM? (ch. 11)

Part C: Calculations

6 points, 2 per question, about 15 min

C1 How many parameters does a fully connected layer with 100 inputs and 10 outputs have? (ch. 2)

C2 How many weights (no bias) does a 1×1 convolution from 256 to 64 channels have? (ch. 5)

C3 A dilated 3×3 convolution has dilation rate 2. What is the span it covers? (ch. 5)